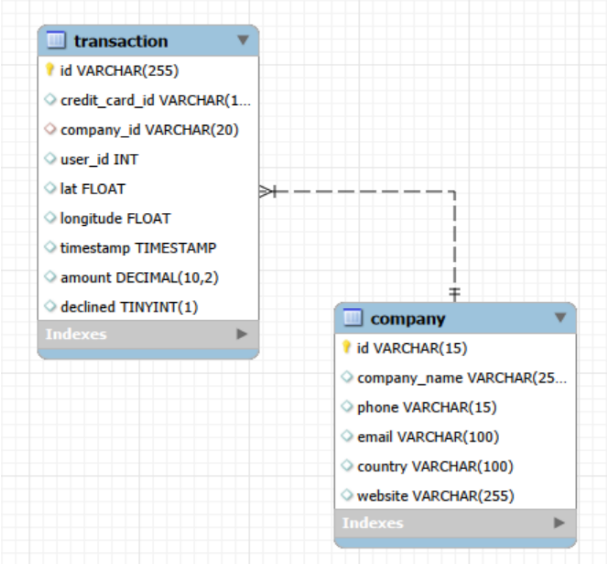
# Nivell 1

## Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.




L'esquema conté dues taules: *transaction* i *company*, ambdues amb una clau primària (PK) anomenada *id*.

Les dues taules estan relacionades mitjançant una clau forana (FK), *company_id*, de la taula *transaction*, que fa referència a *company.id*. Es tracta d'una relació de tipus 1, ja que una companyia pot realitzar diverses transaccions, però cada transacció està associada a una única companyia.

La taula *company* conté les dades identificatives de les companyies (nom, telèfon, email, país i web), mentre que *transaction* recull les dades de cada transacció (targeta de crèdit, companyia, usuari, ubicació, data i hora, import i si ha estat declinada o no).

La línia discontínua del diagrama representa la relació entre les dues taules, i els símbols dels extrems indiquen que una companyia pot estar relacionada amb múltiples transaccions.

## Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes.

```
8   #Llistat dels països que estan generant vendes.
9   SELECT DISTINCT country
10  FROM company c
11  LEFT JOIN transaction t ON t.company_id = c.id
12  WHERE declined = 0;
13
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| country |
|---|
| Netherlands |
| Sweden |
| Ireland |
| United States |
| Belgium |

Result 8

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 10 | 11:38:37 | SELECT DISTINCT country FROM company LEFT JOIN transaction ON transaction.company_id = company.id WHERE decli... | 16 row(s) returned |
| 11 | 11:40:56 | SELECT DISTINCT country FROM company c LEFT JOIN transaction t ON t.company_id = c.id WHERE declined = 0 | 16 row(s) returned |

- Des de quants països es generen les vendes.

```
12  SELECT COUNT(DISTINCT country) AS paisos_vendes
13  FROM company c
14  LEFT JOIN transaction t ON t.company_id = c.id
15  WHERE declined = 0;
16
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| paisos_vendes |
|---|
| 15 |

Result 13

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 15 | 11:44:03 | SELECT COUNT(DISTINCT country) AS paisos_vendes FROM company c LEFT JOIN transaction t ON t.company_id = c.id ... | 1 row(s) returned |

- Identifica la companyia amb la mitjana més gran de vendes.

```sql
SELECT c.id, ROUND(AVG(amount), 2) AS mitjana_vendes
FROM company c
JOIN transaction t ON t.company_id = c.id
WHERE declined = 0
GROUP BY c.id
ORDER BY mitjana_vendes DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| id | mitjana_vendes |
|---|---|
| b-2222 | 284.91 |

Result 16

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 19 | 11:45:49 | SELECT c.id, ROUND(AVG(amount), 2) AS mitjana_vendes FROM company c JOIN transaction t ON t.company_id = c.id WH... | 1 row(s) returned |

## Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```sql
#Mostra totes les transaccions realitzades per empreses d'Alemanya.
SELECT *
FROM transaction t
WHERE declined = 0
AND EXISTS (SELECT 1
            FROM company c
            WHERE country = "Germany"
            AND c.id = t.company_id);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

| id | credit_card_id | company_id | user_id | lat | longitude | timestamp | amount | declined |
|---|---|---|---|---|---|---|---|---|
| 00138D3B-206D-4C03-94B7-63A2676EB9B4 | CcS-4899 | b-2222 | 318 | 41.3781 | 12.447 | 2020-03-25 10:43:43 | 426.36 | 0 |
| 0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9 | CcS-5070 | b-2222 | 489 | 41.3814 | 2.18176 | 2020-12-17 18:15:37 | 316.90 | 0 |
| 00201A11-2E62-44C4-941D-198FC8DB77F0 | CcU-3512 | b-2222 | 193 | 55.5704 | -3.65129 | 2021-01-22 23:44:27 | 453.04 | 0 |
| 00235618-0A5C-4D49-9DCB-B3A9405D8923 | CcS-8137 | b-2222 | 3556 | 59.8421 | 18.729 | 2020-09-09 15:43:19 | 263.14 | 0 |
| 005A5A7B-1F1A-4B6C-9B15-1625A78C9C38 | CcS-8998 | b-2222 | 4417 | 41.1591 | -8.63905 | 2024-05-15 09:10:11 | 442.01 | 0 |

transaction 19

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 22 | 11:50:57 | SELECT * FROM transaction t WHERE declined = 0 AND EXISTS (SELECT 1 FROM company c WHERE country = "Germa... | 13269 row(s) returned |

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```sql
SELECT t.company_id, ROUND(AVG(t.amount), 2) AS mitjana_empresa
FROM transaction t
WHERE t.declined = 0
GROUP BY t.company_id
HAVING AVG(t.amount) > (SELECT AVG(t2.amount)
                        FROM transaction t2
                        WHERE t2.declined = 0
)
ORDER BY mitjana_empresa DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_id | mitjana_empresa |
|---|---|
| b-2222 | 284.91 |
| b-2282 | 275.58 |
| b-2422 | 273.57 |
| b-2538 | 272.74 |
| b-2498 | 270.05 |

Result 23

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 38 | 12:03:21 | SELECT t.company_id, ROUND(AVG(t.amount), 2) AS mitjana_empresa FROM transaction t WHERE t.declined = 0 GROUP ... | 48 row(s) returned |

- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```sql
SELECT *
FROM company c
WHERE NOT EXISTS (SELECT *
                  FROM transaction t
                  WHERE t.company_id = c.id
                  AND declined = 0);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | company_name | phone | email | country | website |
|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL |

company 25

Output

Action Output

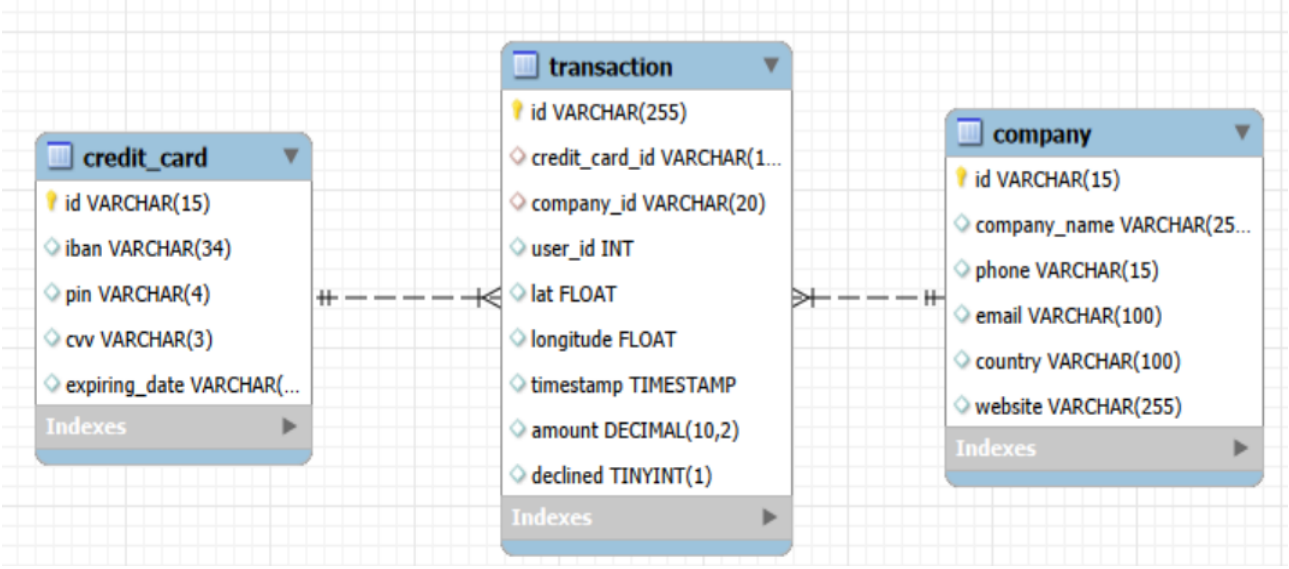
| # | Time | Action | Message |
|---|---|---|---|
| 40 | 12:05:05 | SELECT * FROM company c WHERE NOT EXISTS (SELECT * FROM transaction t WHERE t.company_id = c.i... | 0 row(s) returned |

## Exercici 4

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

Primer creem la taula *credit_card* amb els seus camps (extrets dels inserts), el camp *expiring_date* no és de tipus DATE perquè el seu format és 'mm-dd-yy'.

A continuació executem els INSERTs del document, i per últim fem ALTER TABLE de *transaction* per a definir la clau forània del camp *credit_card_id* al camp *id* de la nova taula. Així queda definida una relació 1:N de *transaction* a *credit_card.*




```sql
CREATE TABLE IF NOT EXISTS credit_card (
    id VARCHAR(15) PRIMARY KEY,
    iban VARCHAR(34),
    pan VARCHAR(20),
    pin VARCHAR(4),
    cvv VARCHAR(3),
    expiring_date VARCHAR(8)
);

ALTER TABLE transaction
ADD CONSTRAINT transaction_fk_2
FOREIGN KEY (credit_card_id)
REFERENCES credit_card(id);
```

## Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

```
59 •  UPDATE credit_card
60    SET iban = "TR323456312213576817699999"
61    WHERE id = "CcU-2938";
62
63 •  SELECT *
64    FROM credit_card
65    WHERE id = "CcU-2938";
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | iban | pan | pin | cvv | expiring_date |
|---|---|---|---|---|---|
| CcU-2938 | TR323456312213576817699999 | 5424465566813633 | 3257 | 984 | 10/30/22 |
| NULL | NULL | NULL | NULL | NULL | NULL |

credit_card 11

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5024 | 11:18:43 | UPDATE credit_card SET iban = "TR323456312213576817699999" WHERE id... | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 |
| 5025 | 11:19:17 | SELECT * FROM credit_card WHERE id = "CcU-2938" | 1 row(s) returned |

## Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

```
Id
108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id
CcU-9999
company_id
b-9999
user_id
9999
lat
829.999
longitude
-117.999
amount
111.11
declined
0
```

Primer afegim una credit_card i una company amb els id's que hi ha al insert de la transaction, i per últim afegim la transaction amb tots els camps que ens donen.

```
68 •  INSERT INTO credit_card(id)
69    VALUES ("CcU-9999");
70
71 •  INSERT INTO company(id)
72    VALUES ("b-9999");
73
74 •  INSERT INTO transaction(id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
75    VALUES ("108B1D1D-5B23-A76C-55EF-C568E49A99DD", "CcU-9999", "b-9999", "9999", "829.999", "-117.999", "111.11", "0");
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5028 | 11:30:31 | INSERT INTO credit_card(id) VALUES ("CcU-9999") | 1 row(s) affected |
| 5029 | 11:30:34 | INSERT INTO company(id) VALUES ("b-9999") | 1 row(s) affected |
| 5030 | 11:30:36 | INSERT INTO transaction(id, credit_card_id, company_id, user_id, lat, longitude, ... | 1 row(s) affected |

```
89 •  SELECT *
90    FROM transaction
91    WHERE id = "108B1D1D-5B23-A76C-55EF-C568E49A99DD";
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | credit_card_id | company_id | user_id | lat | longitude | timestamp | amount | declined |
|---|---|---|---|---|---|---|---|---|
| 108B1D1D-5B23-A76C-55EF-C568E49A99DD | CcU-9999 | b-9999 | 9999 | 829.999 | -117.999 | NULL | 111.11 | 0 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

## Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recorda mostrar el canvi realitzat.

```
78 •  ALTER TABLE credit_card
79    DROP COLUMN pan;
80
81 •  SELECT *
82    FROM credit_card;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

| id | iban | pin | cvv | expiring_date |
|---|---|---|---|---|
| CcS-4857 | XX4857591835292505850771 | 1819 | 467 | 09/27/25 |
| CcS-4858 | XX8581768137002436094025 | 3964 | 817 | 12/28/28 |
| CcS-4859 | XX7826930491423553609370 | 4983 | 277 | 11/26/26 |
| CcS-4860 | XX5559590368835304645299 | 6876 | 661 | 07/27/27 |
| CcS-4861 | XX2035182877195191627307 | 5710 | 398 | 04/25/26 |
| CcS-4862 | XX4774721462463645409758 | 4042 | 174 | 11/27/26 |

credit_card 14 ×

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5030 | 11:30:36 | INSERT INTO transaction(id, credit_card_id, company_id, user_id, lat, longitude, ... | 1 row(s) affected |
| 5031 | 11:35:40 | ALTER TABLE credit_card DROP COLUMN pan | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

## Exercici 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

- american_users.csv
- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:
La taula de products.csv l'utilitzarem més endavant.

```sql
DROP DATABASE IF EXISTS transactions_s2;
CREATE DATABASE IF NOT EXISTS transactions_s2;
USE transactions_s2;

CREATE TABLE IF NOT EXISTS users (
    id INT PRIMARY KEY,
    name VARCHAR(20),
    surname VARCHAR(30),
    phone VARCHAR(20),
    email VARCHAR(60),
    birth_date VARCHAR(15),
    region VARCHAR(20),
    country VARCHAR(20),
    city VARCHAR(30),
    postal_code VARCHAR(15),
    address VARCHAR(50),
    signup_date DATE,
    user_segment VARCHAR(20),
    income_band VARCHAR(10)
);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 53 | 22:03:42 | CREATE TABLE IF NOT EXISTS users ( id INT PRIMARY KEY, name VARCHAR(20), surname VARCHAR(30), phon... | 0 row(s) affected |

```sql
CREATE TABLE IF NOT EXISTS credit_cards (
    id VARCHAR(20) PRIMARY KEY,
    user_id INT REFERENCES users(id),
    iban VARCHAR(34),
    pan VARCHAR(20),
    pin VARCHAR(4),
    cvv VARCHAR(3),
    track1 VARCHAR(100),
    track2 VARCHAR(100),
    expiring_date VARCHAR(8),
    card_type VARCHAR(15),
    card_renewal_flag BOOLEAN
);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 64 | 23:17:59 | CREATE TABLE IF NOT EXISTS credit_cards ( id VARCHAR(20) PRIMARY KEY, user_id INT REFERENCES users(id), ib... | 0 row(s) affected |

```sql
CREATE TABLE IF NOT EXISTS companies (
    company_id VARCHAR(20) PRIMARY KEY,
    company_name VARCHAR(40),
    phone VARCHAR(20),
    email VARCHAR(40),
    country VARCHAR(20),
    website VARCHAR(100),
    merchant_category VARCHAR(20),
    merchant_price_position VARCHAR(20)
);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 66 | 23:19:55 | CREATE TABLE IF NOT EXISTS companies ( company_id VARCHAR(20) PRIMARY KEY, company_name VARCHAR(40), ... | 0 row(s) affected |

```sql
CREATE TABLE IF NOT EXISTS transactions (
    id VARCHAR(40) PRIMARY KEY,
    card_id VARCHAR(20) REFERENCES credit_cards(id),
    business_id VARCHAR(20) REFERENCES companies(company_id),
    timestamp TIMESTAMP,
    amount DECIMAL(10, 2),
    declined BOOLEAN,
    product_ids VARCHAR(50),
    user_id INT REFERENCES users(id),
    lat FLOAT,
    longitude FLOAT,
    discount_amount DECIMAL(10, 2),
    tax_amount DECIMAL(10, 2),
    shipping_amount DECIMAL(10, 2),
    channel VARCHAR(20),
    campaign_id VARCHAR(20),
    device_type VARCHAR(20),
    is_international BOOLEAN,
    decline_reason VARCHAR(50),
    distance_km DECIMAL(10, 2)
);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 68 | 23:21:22 | CREATE TABLE IF NOT EXISTS transactions ( id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(20) REFERENCES cre... | 0 row(s) affected |

```sql
LOAD DATA
LOCAL INFILE 'C:/temp/N1-Ex.8__american_users.csv'
INTO TABLE users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
IGNORE 1 ROWS
(id, name, surname, phone, email, birth_date, country, city, postal_code, address, signup_date, user_segment, income_band)
SET region = 'America';

LOAD DATA
LOCAL INFILE 'C:/temp/N1-Ex.8__european_users.csv'
INTO TABLE users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
IGNORE 1 ROWS
(id, name, surname, phone, email, birth_date, country, city, postal_code, address, signup_date, user_segment, income_band)
SET region = 'Europe';
```

Output

Action Output

| # | Time | Action | Message | D |
|---|---|---|---|---|
| 70 | 23:22:06 | LOAD DATA LOCAL INFILE 'C:/temp/N1-Ex.8__american_users.csv' INTO TABLE users FIELDS TERMINATED BY ',' OPTIO... | 1010 row(s) affected Records: 10... | 0. |
| 71 | 23:22:09 | LOAD DATA LOCAL INFILE 'C:/temp/N1-Ex.8__european_users.csv' INTO TABLE users FIELDS TERMINATED BY ',' OPTIO... | 3990 row(s) affected Records: 39... | 0. |

```
187 •  LOAD DATA
188    LOCAL INFILE 'C:/temp/N1-Ex.8__credit_cards.csv'
189    INTO TABLE credit_cards
190    FIELDS TERMINATED BY ','
191    OPTIONALLY ENCLOSED BY '"'
192    IGNORE 1 ROWS;
193
194 •  LOAD DATA
195    LOCAL INFILE 'C:/temp/N1-Ex.8__companies.csv'
196    INTO TABLE companies
197    FIELDS TERMINATED BY ','
198    OPTIONALLY ENCLOSED BY '"'
199    IGNORE 1 ROWS;
200
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 72 | 23:22:50 | LOAD DATA LOCAL INFILE 'C:/temp/N1-Ex.8__credit_cards.csv' INTO TABLE credit_cards FIELDS TERMINATED BY ',' OP... | 5000 row(s) affected Records: 50... |
| 73 | 23:22:53 | LOAD DATA LOCAL INFILE 'C:/temp/N1-Ex.8__companies.csv' INTO TABLE companies FIELDS TERMINATED BY ',' OPTIO... | 100 row(s) affected Records: 100 ... |

```
201 •  LOAD DATA
202    LOCAL INFILE 'C:/temp/N1-Ex.8__transactions.csv'
203    INTO TABLE transactions
204    FIELDS TERMINATED BY ';'
205    OPTIONALLY ENCLOSED BY '"'
206    IGNORE 1 ROWS;
207
208 •  SELECT * FROM users;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

| id | name | surname | phone | email | birth_date | region | country | city | postal_code | address |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Zeus | Gamble | 1-282-581-0551 | interdum.enim@protonmail.edu | Nov 17, 1985 | America | United States | New York | 10001 | 348-7818 |
| 2 | Garrett | Mcconnell | (718) 257-2412 | integer.vitae.nibh@protonmail.org | Aug 23, 1992 | America | United States | Philadelphia | 19101 | 903 Sit Av |
| 3 | Ciaran | Harrison | (522) 598-1365 | interdum.feugiat@aol.org | Apr 29, 1998 | America | United States | Houston | 77001 | 736-2063 |
| 4 | Howard | Stafford | 1-411-740-3269 | ornare.egestas@icloud.edu | Feb 18, 1989 | America | United States | Phoenix | 85001 | Ap #545- |

users 32

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 80 | 23:29:51 | LOAD DATA LOCAL INFILE 'C:/temp/N1-Ex.8__transactions.csv' INTO TABLE transactions FIELDS TERMINATED BY ';' OPT... | 100000 row(s) affected Records |
| 81 | 23:30:06 | SELECT * FROM users | 5000 row(s) returned |

```
209 •  SELECT * FROM credit_cards;
210
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

| id | user_id | iban | pan | pin | cvv | track1 | track2 |
|---|---|---|---|---|---|---|---|
| CcS-4857 | 276 | ES4156794166769687116992O8 | 2314242385113924 | 1819 | 467 | %B2314242385113924^LWCBUDLWCBUD^22... | %B2314242385111 |
| CcS-4858 | 277 | NL25205303760985126224224 4 | 6582720299715533 | 3964 | 817 | %B6582720299715533^TIQMVITIQMVI^2404... | %B6582720299715 |
| CcS-4859 | 278 | SE7200843185843647714972 62 | 8861684536289642 | 4983 | 277 | %B8861684536289642^COFBGDCOFBGD^280... | %B8861684536289 |
| CcS-4860 | 279 | ES1655740780453143652834 27 | 2481155515498459 | 6876 | 661 | %B2481155515498459^TIUJTUTIUJTU^31040... | %B2481155515498 |

credit_cards 29

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 75 | 23:27:55 | SELECT * FROM credit_cards | 5000 row(s) returned |

```
210 •  SELECT * FROM companies;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| company_id | company_name | phone | email | country | website | me |
|---|---|---|---|---|---|---|
| b-2222 | Ac Fermentum Incorporated | +64 601 244 6320 | donec.porttitor.tellus@yahoo.net | Germany | https://www.ac-fermentum-incorporated.com | Bea |
| b-2226 | Magna A Neque Industries | +33 823 560 3779 | risus.donec.nibh@icloud.org | Australia | https://www.magna-a-neque-industries.com | Trav |
| b-2230 | Fusce Corp. | +29 657 646 4324 | risus@protonmail.edu | United States | https://www.fusce-corp.com | Hom |
| b-2234 | Convallis In Incorporated | +6 693 982 1255 | mauris.ut@aol.couk | Germany | https://www.convallis-in-incorporated.com | Spo |

companies 30 ×

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 77 | 23:28:47 | SELECT * FROM companies | 100 row(s) returned |

```
211 •  SELECT * FROM transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

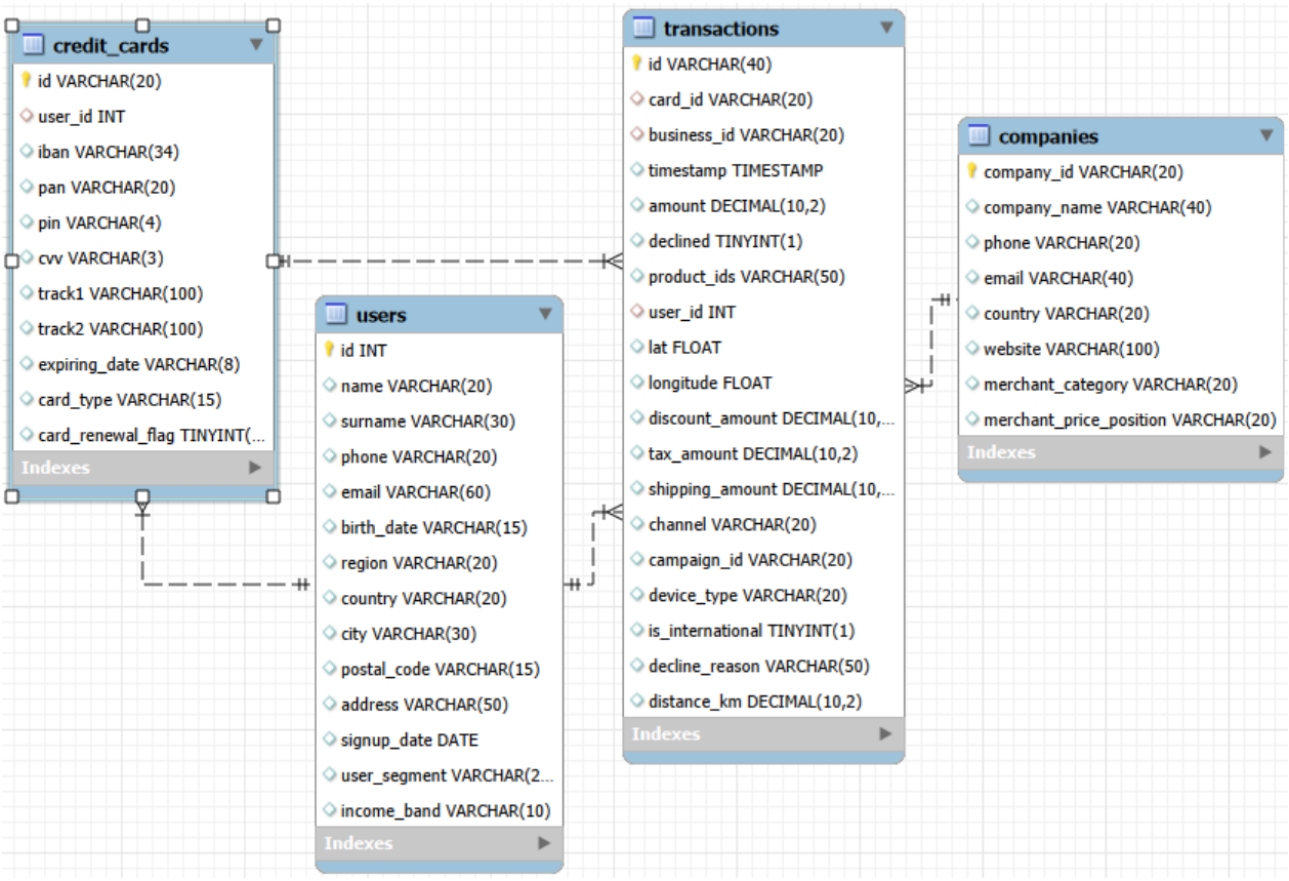
| id | card_id | business_id | timestamp | amount | declined | product_ids | user_id | lat | longitude | disc |
|---|---|---|---|---|---|---|---|---|---|---|
| 00043A49-2949-494B-A5DD-A5BAE3BB19DD | CcS-9294 | b-2458 | 2024-08-28 07:16:46 | 433.51 | 0 | 16, 26, 97, 87 | 4713 | 46.1999 | 1.43554 | 34.1 |
| 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD | CcS-5019 | b-2370 | 2016-12-21 20:07:18 | 172.56 | 0 | 66, 69, 87 | 438 | 41.5972 | 12.2218 | 19.9 |
| 00045D6B-ED2E-4F2F-8186-CEE074D875D0 | CcS-6699 | b-2390 | 2020-07-14 15:37:45 | 332.38 | 0 | 30, 11, 16, 81 | 2118 | 29.7573 | -95.3796 | 16.1 |
| 000481C3-1C26-4FEF-83A0-4CD0EB004BBD | CcS-6696 | b-2230 | 2017-09-04 19:44:53 | 175.00 | 0 | 72 | 2115 | 53.5489 | -113.503 | 1.05 |

transactions 33 ×

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 82 | 23:30:48 | SELECT * FROM transactions | 100000 row(s) returned |




L'esquema conté quatre taules: *users*, *credit_cards*, *companies* i *transactions*, cadascuna amb una PK (*id* o *company_id*).

Les taules estan relacionades mitjançant claus foranes (FK), amb relacions de tipus 1:N:

- *users* → *credit_cards*: un usuari pot tenir diverses targetes de crèdit.
- *users* → *transactions*: un usuari pot realitzar diverses transaccions.
- *credit_cards* → *transactions*: una targeta pot utilitzar-se en diverses transaccions.
- *companies* → *transactions*: una companyia pot tenir diverses transaccions.

La taula *users* conté les dades personals i de contacte dels usuaris (nom, cognoms, telèfon, email, adreça, país, regió...). Hem afegit el camp *region*, al qual hem assignat el valor *Europe* o *America* segons la procedència dels usuaris, ja que les dades provenen de dos arxius CSV diferents.

La taula *credit_cards* guarda la informació de les targetes bancàries dels usuaris.

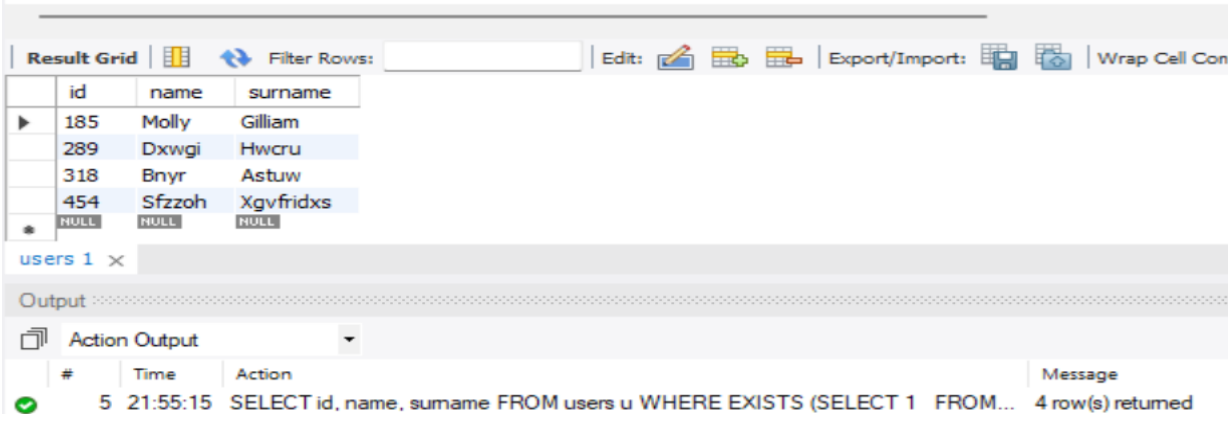
La taula *companies* conté les dades identificatives de les empreses (nom, telèfon, email, país, web...), i *transactions* recull les dades de cada transacció (usuari, targeta, empresa, data, import, productes, descomptes, impostos, despeses d'enviament i si ha estat declinada o no).

Les línies discontínues del diagrama representen les relacions entre les taules, i els símbols dels extrems indiquen si es tracta d'un o de diversos registres.

## Exercici 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```sql
### EXERCICI 9 ###
SELECT id, name, surname
FROM users u
WHERE EXISTS (SELECT 1
              FROM transactions t
              WHERE t.user_id = u.id
              AND t.declined = 0
              GROUP BY t.user_id
              HAVING COUNT(t.id) > 80);
```



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Con

| id | name | surname |
|---|---|---|
| 185 | Molly | Gilliam |
| 289 | Dxwgi | Hwcru |
| 318 | Bnyr | Astuw |
| 454 | Sfzzoh | Xgvfridxs |
| NULL | NULL | NULL |

users 1

Output

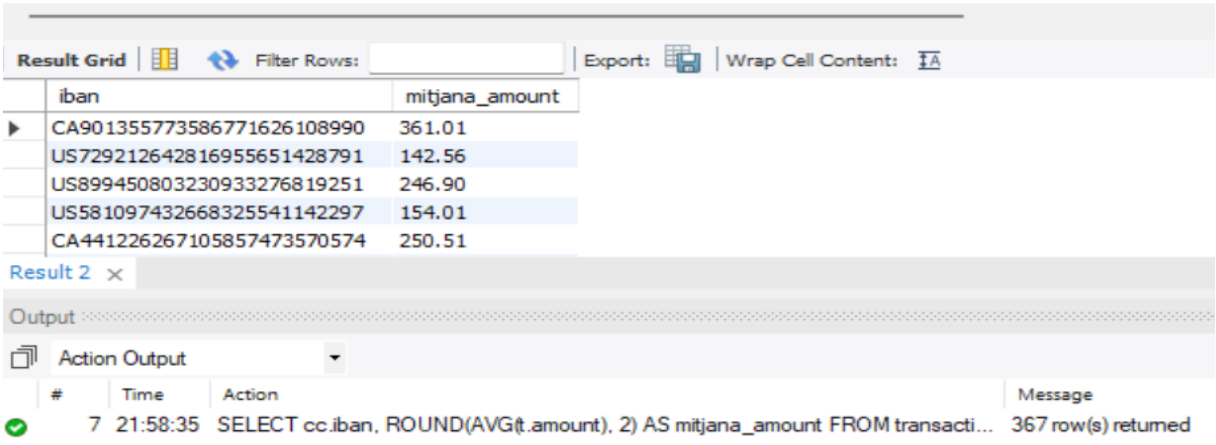
Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5 | 21:55:15 | SELECT id, name, surname FROM users u WHERE EXISTS (SELECT 1 FROM... | 4 row(s) returned |

## Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```sql
SELECT cc.iban, ROUND(AVG(t.amount), 2) AS mitjana_amount
FROM transactions t
JOIN credit_cards cc ON cc.id = t.card_id
WHERE t.business_id = (SELECT c.company_id
                       FROM companies c
                       WHERE c.company_name = 'Donec Ltd')
AND t.declined = 0
GROUP BY cc.iban;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| iban | mitjana_amount |
|---|---|
| CA90135577358677162610899O | 361.01 |
| US72921264281695565142879l | 142.56 |
| US89945080323093327681925l | 246.90 |
| US58109743266832554114229 7 | 154.01 |
| CA44122626710585747357057 4 | 250.51 |

Result 2

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 7 | 21:58:35 | SELECT cc.iban, ROUND(AVG(t.amount), 2) AS mitjana_amount FROM transacti... | 367 row(s) returned |

# Nivell 2

## Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
261 •   SELECT DATE(t.timestamp) AS data_transaccio, SUM(t.amount) AS total_vendes
262     FROM transactions t
263     WHERE t.declined = 0
264     GROUP BY DATE(t.timestamp)
265     ORDER BY total_vendes DESC
266     LIMIT 5;
267
268     ### EXERCICI 2 ###
269 •   SELECT company_name, phone, country, DATE(timestamp) AS data, amount
270     FROM companies
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| data_transaccio | total_vendes |
|---|---|
| 2023-02-20 | 15728.40 |
| 2019-11-18 | 15447.89 |
| 2022-12-13 | 14944.00 |
| 2019-03-18 | 14896.47 |
| 2017-12-20 | 14367.96 |

Result 5

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 10 | 22:01:37 | SELECT DATE(t.timestamp) AS data_transaccio, SUM(t.amount) AS total_vende... | 5 row(s) returned |

## Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
269 •  SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS data, t.amount
270    FROM companies c
271    JOIN transactions t ON t.business_id = c.company_id
272    WHERE t.amount BETWEEN 350 AND 400
273    AND DATE(t.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
274    AND t.declined = 0
275    ORDER BY t.amount DESC;
276
277    ### EXERCICI 3 ###
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_name | phone | country | data | amount |
|---|---|---|---|---|
| Fringilla LLC | +1 150 784 1716 | New Zealand | 2015-04-29 | 390.33 |
| Auctor Mauris Vel LLP | +32 133 447 8239 | United States | 2024-03-13 | 383.58 |
| Mauris Institute | +75 473 619 9472 | Sweden | 2015-04-29 | 369.20 |
| Pede Cum Ltd | +26 625 798 6126 | Norway | 2018-07-20 | 358.64 |
| Netus Et Malesuada Ltd | +52 818 402 1638 | Netherlands | 2024-03-13 | 352.21 |

Result 17

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 39 | 23:52:15 | SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS data, t.am... | 6 row(s) returned |

## Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```
277 •  SELECT c.company_id, COUNT(t.id) AS num_transaccions,
278        CASE
279            WHEN COUNT(t.id) >= 400 THEN '400 o més'
280            ELSE 'Menys de 400'
281        END AS categoria
282    FROM companies c
283    LEFT JOIN transactions t ON t.business_id = c.company_id AND t.declined = 0
284    GROUP BY c.company_id
285    ORDER BY num_transaccions DESC;
286
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_id | num_transaccions | categoria |
|---|---|---|
| b-2222 | 2387 | 400 o més |
| b-2302 | 1582 | 400 o més |
| b-2330 | 1578 | 400 o més |
| b-2366 | 1575 | 400 o més |
| b-2614 | 1573 | 400 o més |

Result 9

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 14 | 22:05:07 | SELECT c.company_id, COUNT(t.id) AS num_transaccions, CASE WHEN ... | 100 row(s) returned |

## Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```
245 •  DELETE FROM transactions
246    WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD";
247
248 •  SELECT *
249    FROM transactions
250    WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD";
251
252
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conten

| id | card_id | business_id | timestamp | amount | declined | product_ids | user_id | lat | longitude | disco |
|---|---|---|---|---|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

transactions 23

Output

Action Output

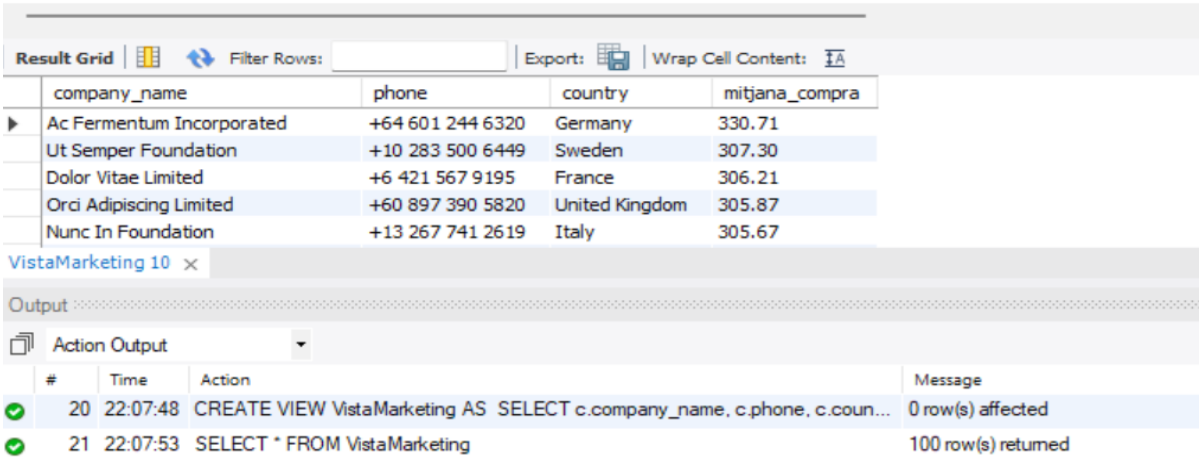
| # | Time | Action | Message |
|---|---|---|---|
| 44 | 17:27:11 | DELETE FROM transactions WHERE id = "000447FE-B650-4DCF-85DE-C7ED0... | 1 row(s) affected |

## Exercici 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```sql
CREATE VIEW VistaMarketing AS
SELECT c.company_name, c.phone, c.country, ROUND(AVG(t.amount), 2) AS mitjana_compra
FROM companies c
LEFT JOIN transactions t ON t.business_id = c.company_id AND t.declined = 0
GROUP BY c.company_id
ORDER BY mitjana_compra DESC;

SELECT * FROM VistaMarketing;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_name | phone | country | mitjana_compra |
|---|---|---|---|
| Ac Fermentum Incorporated | +64 601 244 6320 | Germany | 330.71 |
| Ut Semper Foundation | +10 283 500 6449 | Sweden | 307.30 |
| Dolor Vitae Limited | +6 421 567 9195 | France | 306.21 |
| Orci Adipiscing Limited | +60 897 390 5820 | United Kingdom | 305.87 |
| Nunc In Foundation | +13 267 741 2619 | Italy | 305.67 |

VistaMarketing 10

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 20 | 22:07:48 | CREATE VIEW VistaMarketing AS SELECT c.company_name, c.phone, c.coun... | 0 row(s) affected |
| 21 | 22:07:53 | SELECT * FROM VistaMarketing | 100 row(s) returned |

# Nivell 3

## Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:
? Quantes targetes estan actives?

```sql
CREATE TABLE IF NOT EXISTS credit_card_estat AS
SELECT cc.id,
CASE
        WHEN COUNT(tn.id) = 3 AND SUM(tn.declined = 1) = 3 THEN 'Inactiva'
        ELSE 'Activa'
    END AS estat
FROM credit_cards cc
LEFT JOIN (SELECT t.id, t.card_id, t.declined, t.timestamp,
            ROW_NUMBER() OVER (
                PARTITION BY t.card_id
                ORDER BY t.timestamp DESC, t.id DESC
            ) AS num_transaccio
            FROM transactions t
) tn
ON tn.card_id = cc.id AND tn.num_transaccio <= 3
GROUP BY cc.id;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 25 | 22:10:59 | CREATE TABLE IF NOT EXISTS credit_card_estat AS SELECT cc.id, CASE ... | 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0 |

```sql
SELECT COUNT(*) AS targetes_actives
FROM credit_card_estat cce
WHERE cce.estat = 'Activa';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| targetes_actives |
|---|
| 5000 |

Result 12

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 26 | 22:11:10 | SELECT COUNT(*) AS targetes_actives FROM credit_card_estat WHERE estat ... | 1 row(s) returned |

## Exercici 2

Crea una taula amb la qual poguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podíem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta:
? Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

```sql
CREATE TABLE IF NOT EXISTS products (
    id INT PRIMARY KEY,
    product_name VARCHAR(80),
    price DECIMAL(10,2),
    currency VARCHAR(3) DEFAULT 'USD',
    colour VARCHAR(7),
    weight DECIMAL(10,1),
    warehouse_id VARCHAR(10),
    category VARCHAR(30),
    brand VARCHAR(30),
    cost DECIMAL(10,2),
    launch_date DATE
);
```

```sql
LOAD DATA
LOCAL INFILE 'C:/temp/N1-Ex.8__products.csv'
INTO TABLE products
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
IGNORE 1 ROWS
(id, product_name, @price, colour, weight, warehouse_id, category, brand, @cost, launch_date)
SET price = CAST(REPLACE(@price, '$', '') AS DECIMAL(10,2)),
    cost = CAST(REPLACE(@cost, '$', '') AS DECIMAL(10,2)),
    currency = 'USD';

SELECT * FROM products;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | product_name | price | currency | colour | weight | warehouse_id | category | brand | cost | launch_date |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Direwolf Stannis | 161.11 | USD | #7c7c7c | 1.0 | WH-4 | Sports | Runway Pro | 63.84 | 2014-11-01 |
| 2 | Tarly Stark | 9.24 | USD | #919191 | 2.0 | WH-3 | Gaming | Questbyte | 4.74 | 2013-02-07 |
| 3 | duel tourney Lannister | 171.13 | USD | #d8d8d8 | 1.5 | WH-2 | Food | Daily Pantry | 107.07 | 2013-08-18 |
| 4 | warden south duel | 71.89 | USD | #111111 | 3.0 | WH-1 | Books | NorthPress | 39.17 | 2013-12-31 |
| 5 | skywalker ewok | 171.22 | USD | #dbdbdb | 3.2 | WH-0 | Home | NordHaus | 83.69 | 2014-07-17 |

products 13

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 30 | 22:30:16 | CREATE TABLE IF NOT EXISTS products ( id INT PRIMARY KEY, product_n... | 0 row(s) affected |
| 31 | 22:31:38 | LOAD DATA LOCAL INFILE 'C:/temp/N1-Ex.8__products.csv' INTO TABLE pro... | 100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0 |
| 32 | 22:32:10 | SELECT * FROM products | 100 row(s) returned |

```sql
CREATE TABLE IF NOT EXISTS products_transactions (
    id_transaction VARCHAR(40),
    id_producte INT,
    PRIMARY KEY (id_transaction, id_producte),
    FOREIGN KEY (id_transaction) REFERENCES transactions(id),
    FOREIGN KEY (id_producte) REFERENCES products(id)
);

INSERT INTO products_transactions (id_transaction, id_producte)
SELECT t.id AS id_transaction, jt.id_product
FROM transactions t
JOIN JSON_TABLE(
    CONCAT('[', t.product_ids, ']'),
    '$[*]' COLUMNS (
        id_product INT PATH '$'
    )
) AS jt;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 33 | 22:33:39 | CREATE TABLE IF NOT EXISTS products_transactions ( id_transaction VAR... | 0 row(s) affected |
| 34 | 22:33:44 | INSERT INTO products_transactions (id_transaction, id_producte) SELECT t.id A... | 253388 row(s) affected Records: 253388 Duplicates: 0 Warnings: 0 |

```sql
SELECT * FROM products_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

| id_transaction | id_producte |
|---|---|
| 001A60EA-DC9C-4E5A-9460-6628B100E7E1 | 1 |
| 0032F0BB-BBE6-4AA5-B5EE-EEAD533C0C48 | 1 |
| 00342381-503D-422D-85AB-F2D4FFAAD4C7 | 1 |
| 004C0A80-E537-46D8-BE44-343D2176DF15 | 1 |
| 004D1DB5-B2CB-4460-98B6-31C42CA96E5F | 1 |

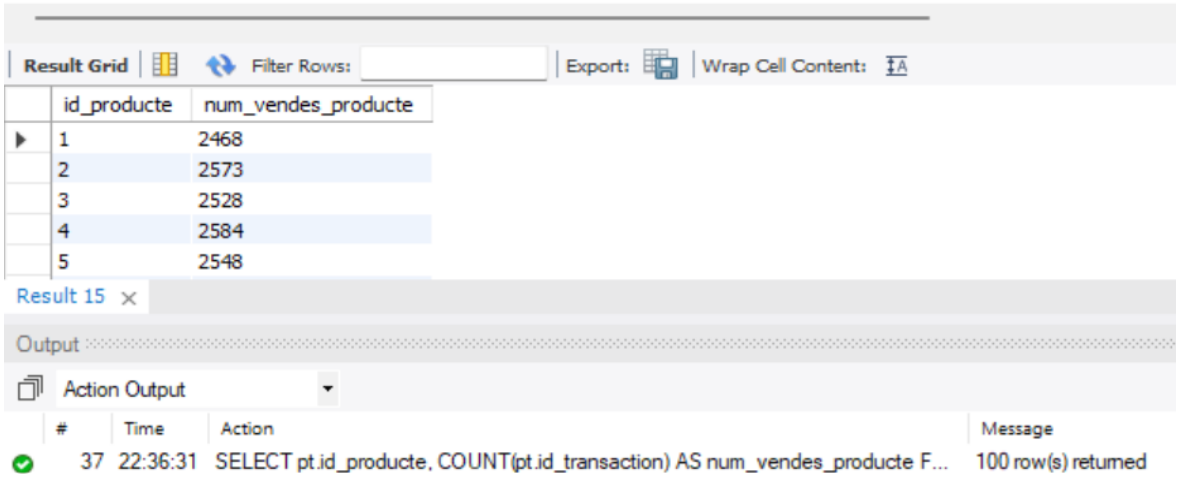
products_transactions 14

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 35 | 22:35:22 | SELECT * FROM products_transactions | 253388 row(s) returned |

```
377 •   SELECT pt.id_producte, COUNT(pt.id_transaction) AS num_vendes_producte
378     FROM products_transactions pt
379     GROUP BY pt.id_producte
380     ORDER BY pt.id_producte;
381
```




A la taula *products* hem definit els camps *price* i *cost* com a DECIMAL(10,2), ja que contenen imports monetaris. Com que al fitxer CSV aquests valors inclouen el símbol $, durant la importació hem utilitzat REPLACE() i CAST() per eliminar el símbol i convertir-los a valors numèrics.

També hem afegit el camp *currency*, amb el valor per defecte *USD*, per identificar la moneda dels imports i permetre treballar amb altres monedes en el futur sense perdre aquesta informació.